

HARIS WAHEED BHATTI

BIM Specialist | Information Manager (ISO 19650)

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PROFILE

Civil engineer with a European Master's in Building Information Modelling, combining strong civil engineering fundamentals with digital coordination skills. Owned a federated model's clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650, and modelled civil structures from 2D drawings into 3D during an internship.

PROJECTS

Federated BIM Model – Pakistani-Themed Expo, Guimarães | IFC, Autodesk Construction Cloud Oct 2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

ifctoneo4j – Python package (PyPI) | Python, IfcOpenShell, Neo4j, Cypher 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

BIMGraphExplorer – LLM agent for IFC question answering | Python, MCP, Neo4j 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.
- Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge; 23 IFC datasets, 367 questions, 15 projects). Overall accuracy 72.0% against the baseline's 82.3%, but reached source-transparency parity (99.7% versus 100%) and outperformed it on 5 of 23 datasets, by up to 24 points on one.

EDUCATION

MSc, European Master in BIM (BIM A+) | UL FGG, Ljubljana, Slovenia Oct 2025 – Jul 2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. hdl.handle.net/20.500.12556/RUL-185169
- Applied coursework spanning the BIM delivery chain: IFC schema auditing and model remediation, parametric façade optimisation in Dynamo, process modelling and workflow improvement, 5D quantity take-off and cost estimation, and scan-to-BIM from point-cloud capture.

BE Civil Engineering | NUST, Islamabad, Pakistan. Final grade 3.25/4.00 Sep 2020 – Jun 2024

EXPERIENCE

Civil Engineering Intern | Fatima Fertilizer Company, Pakistan July 2023

- Modelled a 6,804 m² rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

Contract Management Intern | Engineering and Management Services, Pakistan June 2023

- Reviewed contract reports and arbitration documentation under FIDIC conditions; presented findings to the internal legal team.

Research Intern (remote) | Southern Illinois University Edwardsville, USA July 2023

- Synthesised academic literature to support a faculty-mentored research proposal.

SKILLS

BIM & AEC: Dynamo (parametric design), Revit, clash detection, federated models, IFC (IFC2X3, IFC4, IFC4X3), ISO 19650, Employer's Information Requirements, Autodesk Construction Cloud, IfcOpenShell, Linked Building Data ontologies

Programming & Data: Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git

AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking

Languages: English (C1)